



YOAV DABAS — ARCHITECT & RESEARCHER

Architecture of Things

Bringing processes and systems together into something new — a technology, a method, a product, a function.

Portfolio · dabasy8.github.io/portfolio · dabasy8@gmail.com

ABOUT

I'm an architect and entrepreneur (M.Arch, Technion — specialization track in Computational Design and Manufacturing) working at the intersection of computational design, material research, and robotic fabrication. WoodenWood, a collaborative project with my partners at the Disrupt.Design Lab, received the Design Educates Gold Prize and the Fuorisalone Award, and was nominated for the Dezeen Awards.

Today I work as an architect at Bar Orian Architects while continuing research in additive manufacturing. I also served as a Captain in the Armored Corps, commanding soldiers and officers in active duty and reserve.

AWARDS

Gold Prize, Design Educates Awards (2023) — Emerging Designers category · WoodenWood (collaborative)

Winner, Fuorisalone Award (2023) — Milan Design Week · WoodenWood (collaborative)

Nominee, Dezeen Awards (2023) — Emerging Architect/Design category · WoodenWood (collaborative)

Nominee, Dori Award (2024) — Social Hub at Gilat Junction

CONTENTS

ADDITIVE MANUFACTURING & ARCHITECTURE

Fertile City

MyBriX

DESIGN STUDIO

Hareshet 17

One and One Third

If Alone, We Better Be Moving

ADDITIVE MANUFACTURING RESEARCH

WoodenWood

COMPUTATIONAL DESIGN

Independent Work

ADDITIVE MANUFACTURING & ARCHITECTURE

Fertile City

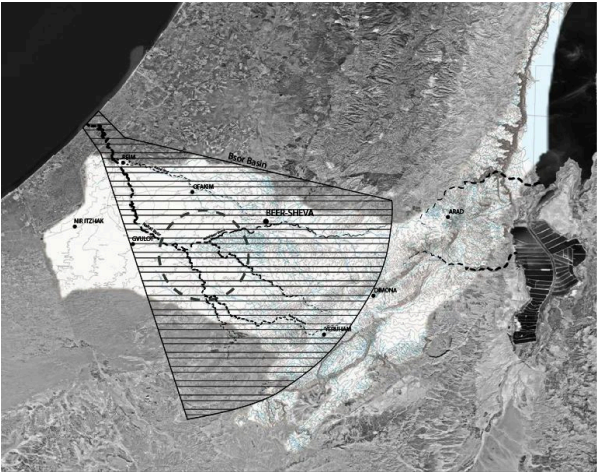
Architecture — 2025 — Technion

Fertile City is a design research project addressing desertification along the Besor Stream in the northern Negev. Reading the site's erosion and drainage patterns as design drivers, the project proposes a water-retention suburb and a wind-diffusing strip to stabilize the soil and restore fertility to the region. A parallel fabrication process develops a soil-based building material — combining local soil, biochar, cellulose, and psyllium — cast and dried into structural components for the city's construction.

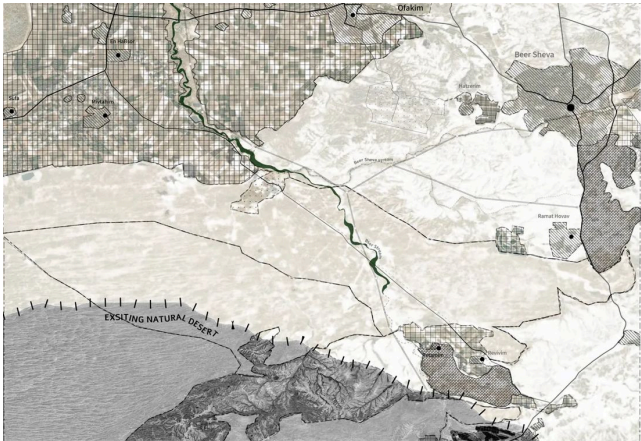


Fertile City — Site & Desertification Process

Mapping the Besor Stream site — its drainage basin, erosion patterns, and the feedback loop driving desertification — as the analytical basis for the design proposal.



Drainage Basin



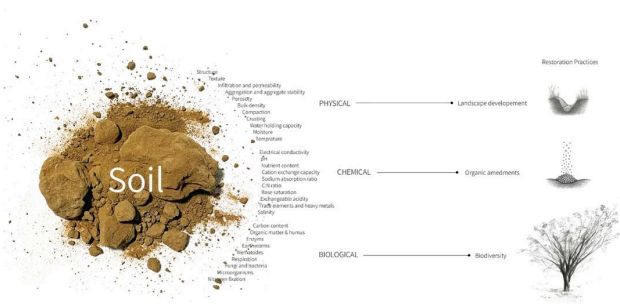
Mapping Erosion



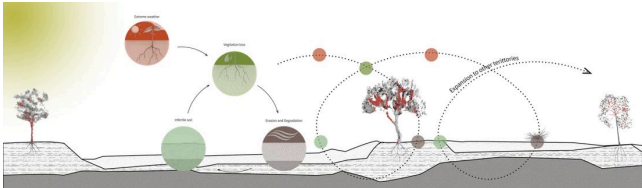
Besor Stream



Lonely Tree



Soil & Restoration



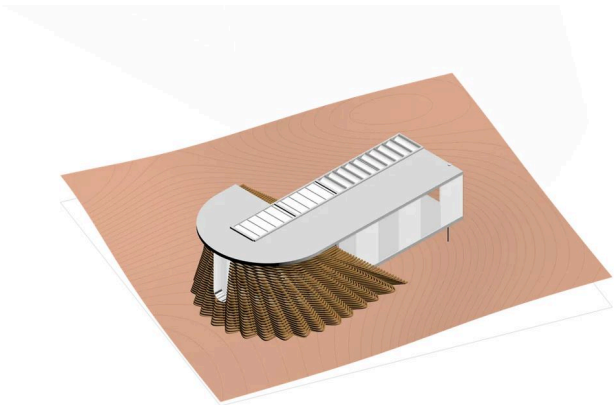
Desertification Feedback Loop

Fertile City — Water Retention Suburb

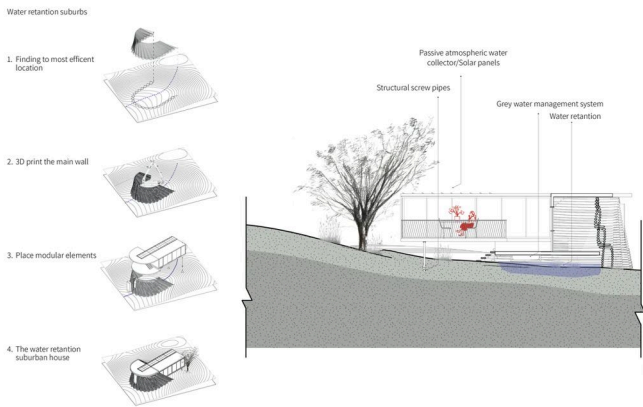
A residential typology designed to capture and retain seasonal stream water, using restorative architecture to slow runoff and rebuild soil moisture across the suburb.



Model



Isometric



Section



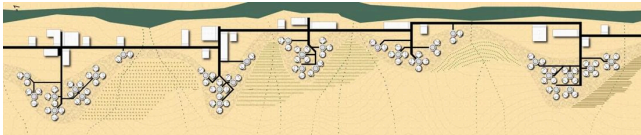
Restorative Architecture

Fertile City — Wind Diffusing Strip

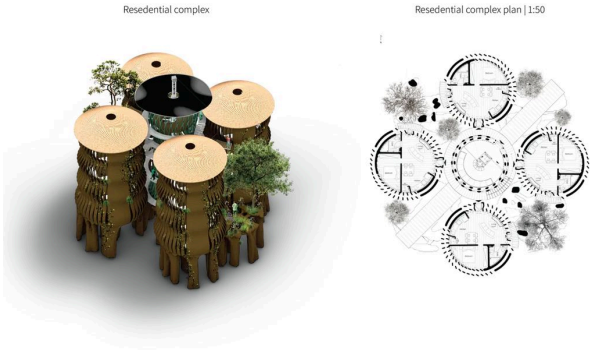
A linear residential complex that breaks and diffuses desert wind at the settlement edge, developed from program and layout through to column, unit, and wall-section detailing.



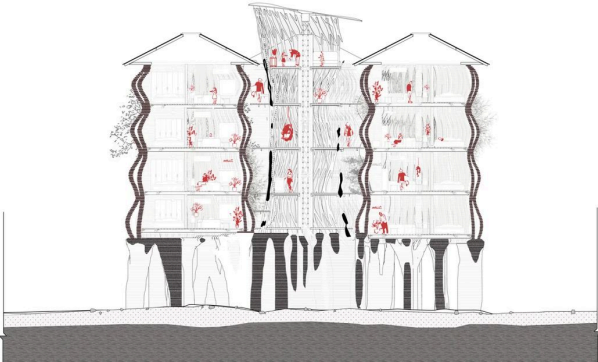
Program



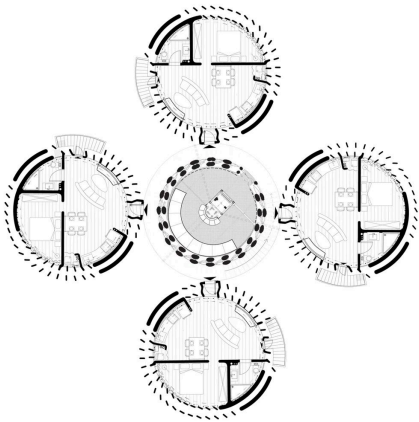
Layout



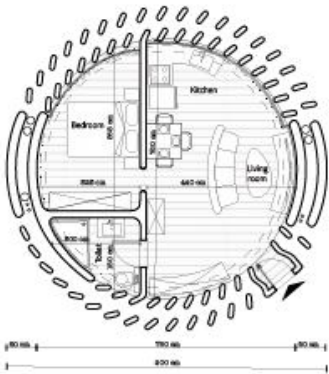
Complex Layout & Plan



Complex Section



Complex Plan



Unit Plan

Fertile City — Fertile City

The completed urban vision — bringing the desertification analysis, water-retention suburb, and wind-diffusing strip together into a single settlement.



Fertile City 1



Fertile City 2

Fertile City — Fabrication Process

Developing a soil-based building material from local loess soil, biochar, cellulose, and psyllium — mixed, cast, and dried into structural components for the city's construction.



Collecting Soil



Loess Soil



Biochar & Cellulose



Soil & Psyllium



Soil Mixture



Biochar, Soil & Cellulose

ADDITIVE MANUFACTURING & ARCHITECTURE

MyBriX

Architecture / Material Research — 2022

MyBriX is a material research project developing living, fungus-grown building blocks. Mycelium is cultivated under controlled protocols, combined with 3D-printed forms, and grown into bricks that are catalogued, tested, and assembled into walls — from lab jar to architectural element.

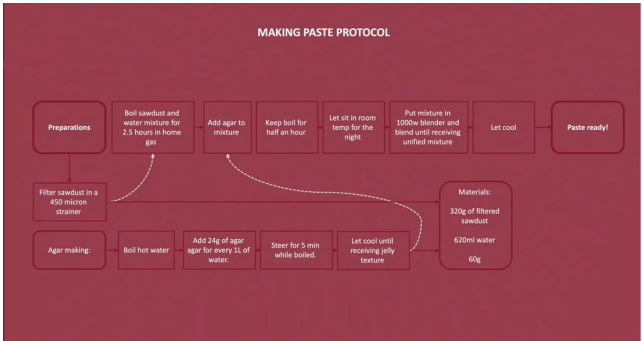


MyBriX — Cultivation

Growing the fungus — testing substrates, jars, and cultivation protocols, and tracking growth over time to find reliable conditions.



Fungus Cultivation



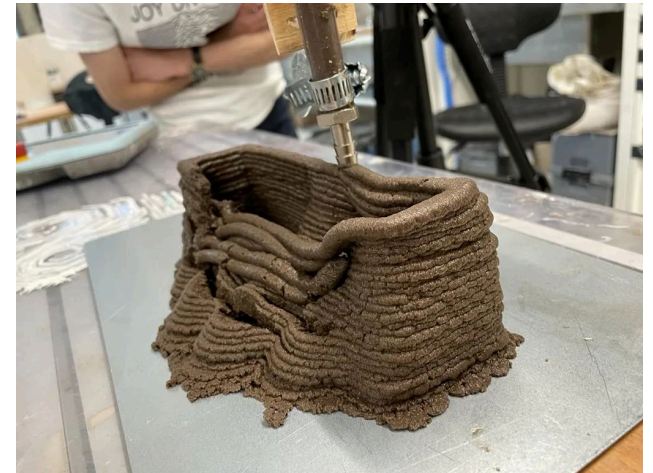
Cultivation Protocol

MyBriX — Printing Process

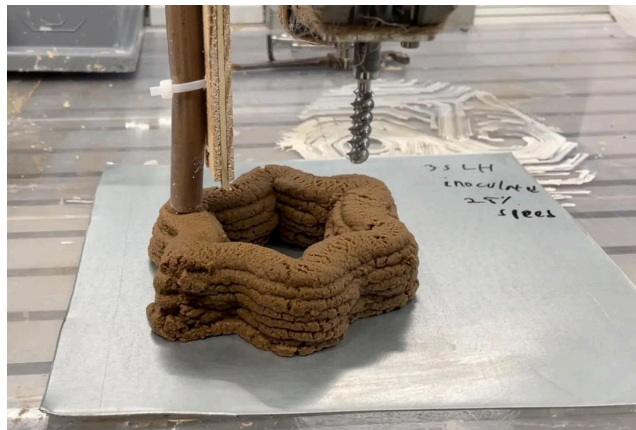
Setting up the printing workflow, printing brick geometries, and learning from failed prints to tune material and toolpath.



Printing Experiment



Failed Print



Printing Brick

MyBriX — Parametric Catalog

A parametric family of brick geometries — varying pattern, porosity, and form — generated to explore the design space of the system.



MyBriX — Wall Assembly

Grown and dried bricks assembled into wall prototypes, testing how individual units come together as an architectural surface.



Dried Brick Wall



DESIGN STUDIO

Hareshet 17

Integrative Studio — 2024 — Revit / Lumion / Physical Modeling

An education center combining kindergartens, a library, an art gallery, and a children's psychology center. The program was translated into volumes and composed through physical modeling. The structure is reinforced concrete at ground level and steel from Level 1 upward. Cladding follows a gravitational logic — stone, Dekton, and milky glass, heavy and coarse at the base, light and delicate at the top. Detailed design was developed in Revit.



Hareshet 17 — Concept



Physical Model 1

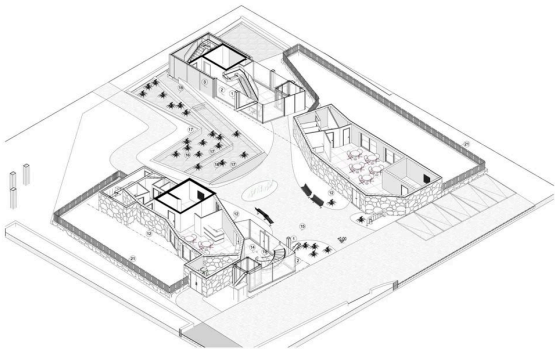


Physical Model 2

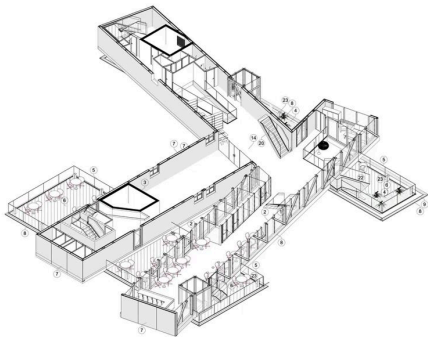
Hareshet 17 — Detailed Design



Axonometric



Ground Floor Axonometric



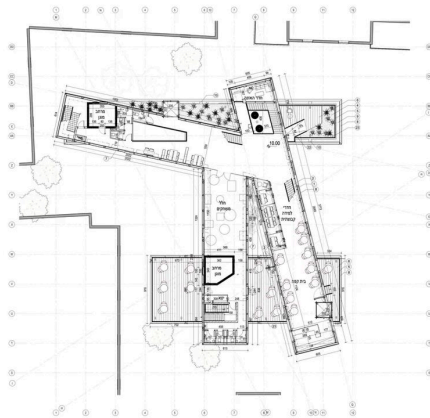
Level 2 Axonometric



East Elevation



Ground Floor Plan



Level 2 Floor Plan

Hareshet 17 — Renders



Coffee View



View from Street



View from Balcony



Top View

DESIGN STUDIO

One and One Third

Parametric Studio — 2021

A building with an ecological façade designed to serve non-human stakeholders alongside its human users. Parametric tools and environmental simulations drive the façade's form, shaping it into habitat and infrastructure for the life around it.

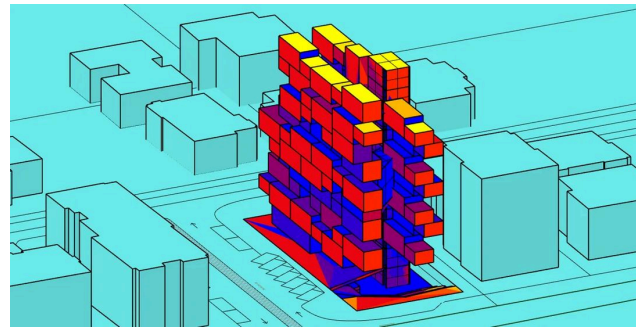


One and One Third — Site & Analysis

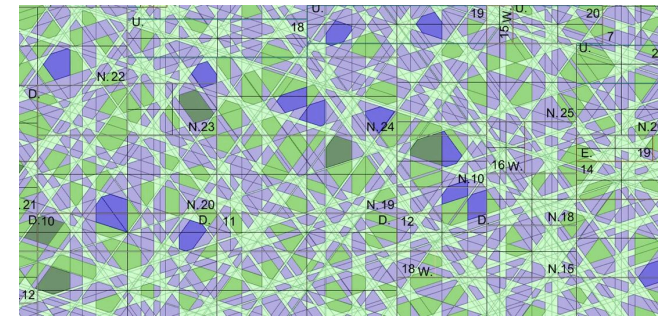
Reading the site and its light conditions, and mapping which species each part of the façade can support.



The Site



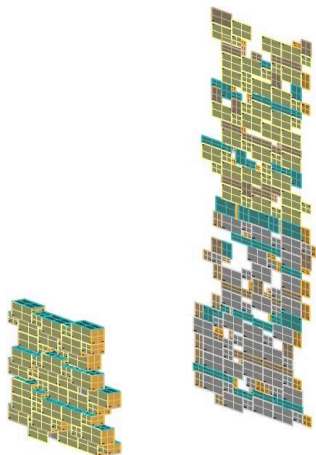
Sunlight Analysis



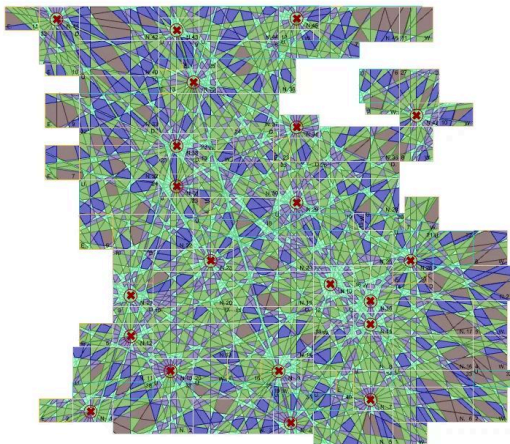
Façade Species Mapping

One and One Third — Façade System

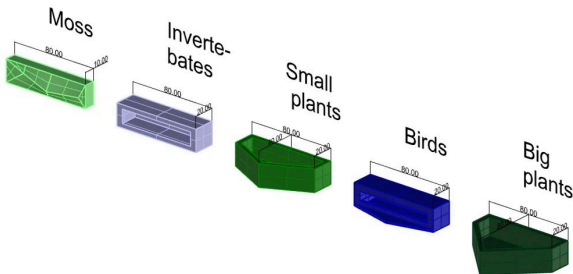
The building volume is unwrapped into a tiled façade, with each tile adjusted to the needs of a specific species — moss, invertebrates, plants and birds.



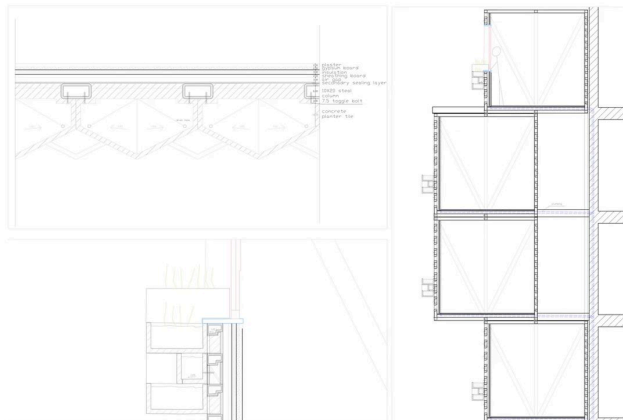
Unwrapping the Volume



Façade Development



Tile Species Adjustments



Tile System Detail



North Elevation — Species Composite

One and One Third — Renders

The building within its restored landscape.



View from the Hill



Render

DESIGN STUDIO

If Alone, We Better Be Moving

Design Studio — Kiryat Yam

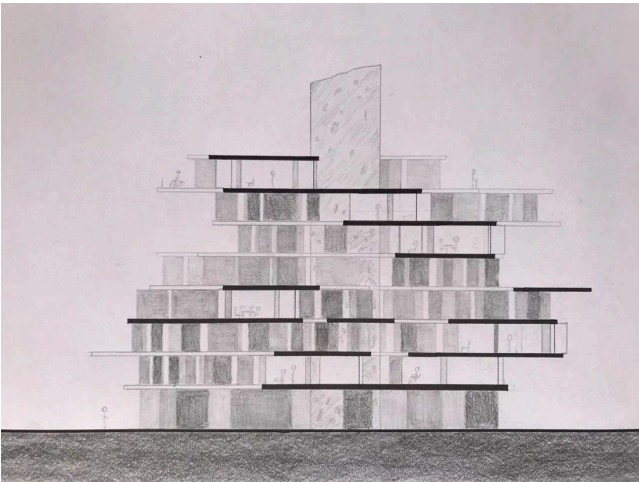
Set on the coast of Kiryat Yam, where wind is the strongest force on site, the design works with the wind rather than against it. Repetitive modules break apart and reconnect to form a dynamic architectural system, open and shifting in response to its environment.



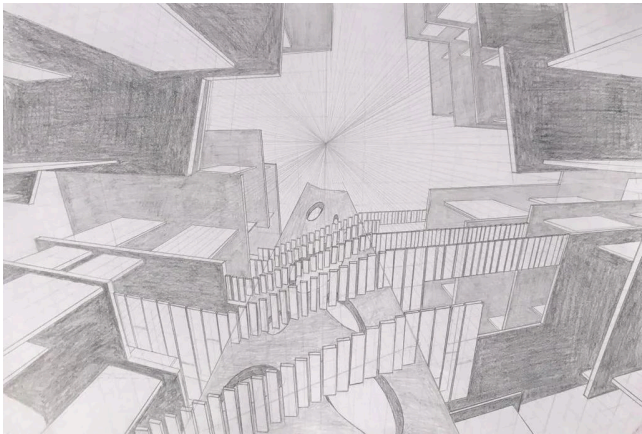
If Alone, We Better Be Moving — Concept & Drawings



Modular Field



Elevation



Perspective Sketch

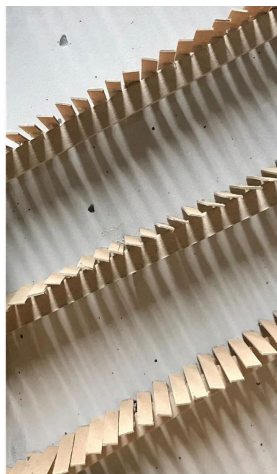
If Alone, We Better Be Moving — Models & Materials



Physical Model



Model Interior



Staircase System



Concrete Sample



Material Study

ADDITIVE MANUFACTURING RESEARCH

WoodenWood

Collaborative Research — 2024 — Robotic Printing

In collaboration with Avi Cohen, Yuval Berger, Alon Nisan & Shany Barath — Disrupt.Design Lab, Technion

Wood waste repurposed into seating through robotic printing and traditional woodworking. A parametric model links stool design, printing toolpaths, and material properties — balancing ergonomics, function, and material distribution — and produces a woven "wooden textile" deposition. Published at CAADRIA 2024; Gold Prize, Design Educates Awards 2023.



WoodenWood — Concept & Process



Wood Texture



The Motivation



Wood Lifecycle — Chair



Wood Lifecycle — Stools

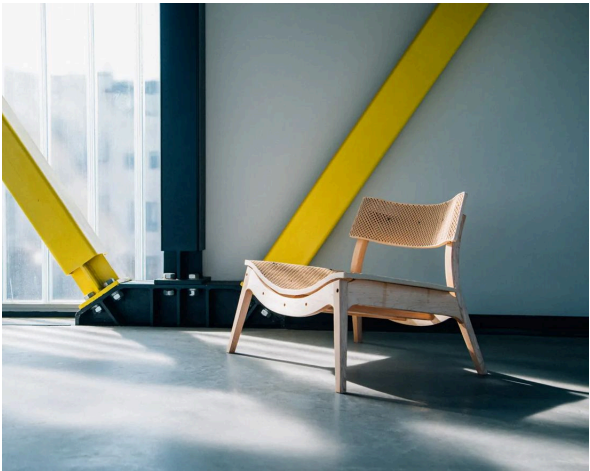


Structure as a Mold

WoodenWood — Stools & Chair



Lounge Chair



Chair in Space



Stools — Woven Seats



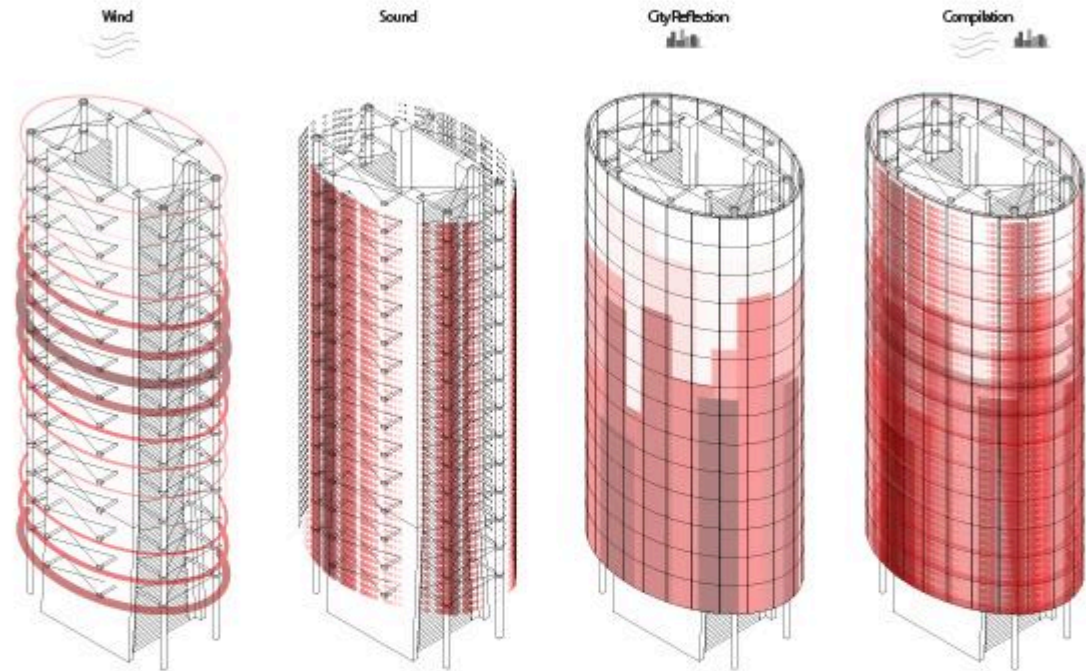
Wood Waste & Woven Texture

COMPUTATIONAL DESIGN

Independent Work

Computational Design

Self-initiated studies in computational design — form-finding, simulation, and structural logic explored through code.



TOWER OF WINDS - RESPONSIVE ARCHITECTURE

YOYO IS KNOWN AS AN ARCHITECT WHO TENDS TO ADDRESS NATURE THROUGH ADVANCED TECHNOLOGY IN HIS PROJECTS. TO HIS PHILOSOPHY, ARCHITECTURE SHOULD ENCOURAGE A BETTER FUTURE THAT COMBINES HIGH TECHNOLOGY AND NATURAL ELEMENTS.

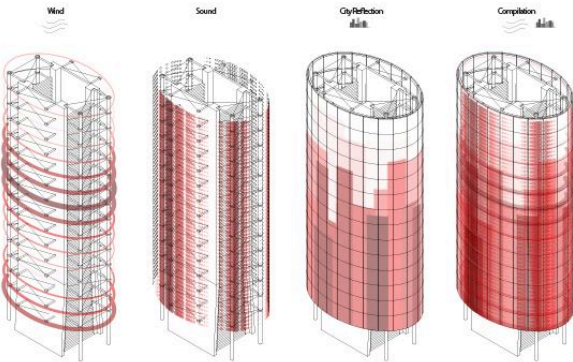
TOWER OF WINDS CREATES A DIALOG WITH ITS SURROUNDINGS. IT HOLDS ELECTRONIC SYSTEMS THAT SENSE EXTERNAL ELEMENTS AND TRANSLATE THEM INTO VISUAL OUTPUTS. THE ELEMENTS THE BUILDING COLLECTS ARE WIND, NOISES AND THE CITY'S OUTLINE. THE WIND POWER AND TENDENCY IS REFLECTED THROUGH ELIPTICAL LED STRIPS THAT WRAPS THE BUILDING IN HORIZONTAL DIRECTION THAT GOES FROM TOP TO BOTTOM. THE NOISES THAT COMES FROM THE CITY ARE TRANSFORMED INTO LIGHTED DOTS. THE DOTS ARE SHOWN THROUGH THE ALUMINIUM PANELS WHO SEEMS TRANSPARENT DURING THE NIGHT. DURING THE DAY THE PANELS ON THE BUILDINGS ENVELOP WORKS LIKE MIRROR AND REFLECT THE CITY'S CONTOUR.

THE FORCES IN THAT BUILDING IS THE WAY IT RESPONDS TO ITS ENVIRONMENT AND KEEP CHANGES ITS FACADE THROUGH THE DAY AND NIGHT AND TO DYNAMICS OF THE CITY.

Independent Work

Tedrisim Israel Institute of Technology | Evaluative Tools in Architecture 2021

Tower of Winds - Forces - Yoram Dabas



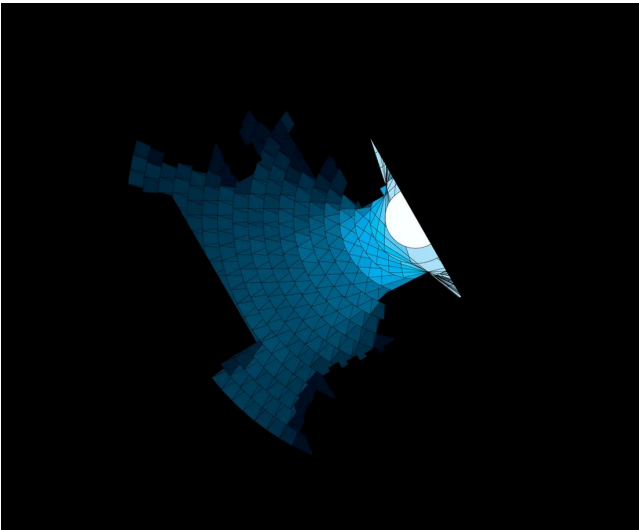
TOWER OF WINDS - RESPONSIVE ARCHITECTURE

TODD ITD KNOWN AS AN ARCHITECT WHO TEND TO ADDRESS THE NATURE THROUGH ADVANCED TECHNOLOGY IN HIS PROJECTS. TO HIS PHILOSOPHY, ARCHITECTURE SHOULD ENCOURAGE A BETTER FUTURE THAT COMBINING HIGH TECHNOLOGY AND NATURAL ELEMENTS.

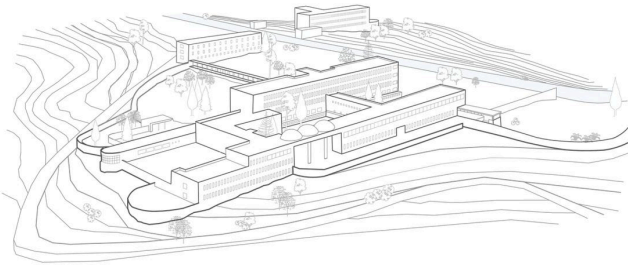
TOWER OF WINDS CREATE A DIALOGUE WITH ITS SOUNDS. IT HOLDS ELECTRONIC SYSTEMS THAT SENSE EXTERNAL ELEMENTS AND TRANSLATE THEM INTO VISUAL OUTPUTS. THE ELEMENTS THE BUILDING COLLECTS ARE WIND, NOISES AND THE CITY'S OUTLINE. THE WIND POWER AND TENDENCY REFLECTED THROUGH ELLIPTICAL LED STRIPS THAT WRAP THE BUILDING IN HORIZONTAL DIRECTION THAT GOES FROM TOP-TO-BOTTOM. THE NOISES THAT COMES FROM THE CITY ARE TRANSFORMED INTO LIGHTED DOTS. THE DOTS ARE SHOWN THROUGH THE ALUMINUM PANELS WHO SEEM TRANSPARENT DURING THE NIGHT. DURING THE DAY THE PANELS ON THE BUILDINGS ENVELOP WORKS LIKE MIRROR AND REFLECT THE CITY'S CONTOUR.

THE FORCES IN THAT BUILDING IS THE WAY IT RESPOND TO ITS ENVIRONMENT AND KEEP CHANGES ITS FACADE THROUGH THE DAY AND NIGHT AND TO DYNAMICS OF THE CITY.

Tower of Winds — Forces Diagram



POV — Panelled Surface



Har Hatzofim — Axonometric Drawing

Yoav Dabas

dabasy8@gmail.com

LinkedIn: www.linkedin.com/in/yoav-dabas/ · X: x.com/DabasYoav · GitHub: github.com/dabasy8

Full portfolio online: dabasy8.github.io/portfolio